

NANGMAGVIK LAKE, NT, Canada

WMO: 719740

Lat: 74.139N

Lon: 119.989W

Elev: 32

StdP: 100.94

Time Zone: -7.00 (NAM)

Period: 96-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-43.0	-41.6	-47.6	0.0	-42.9	-46.1	0.0	-41.4	15.6	-19.5	13.6	-22.4	2.0	150	1.125

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	17.5	13.0	14.6	11.3	11.8	9.4	13.2	17.0	11.4	14.4	9.5	11.7	5.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.9	8.2	14.6	9.2	7.2	13.0	7.7	6.5	11.1	37.2	17.0	32.7	14.4	28.2	11.7	17.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.7	10.3	DB	-44.1	19.9	1.8	2.7	-45.4	21.8	-46.4	23.4	-47.5	24.9	-48.8	26.9	(4)
(5)			WB	-44.2	14.7	1.7	1.9	-45.4	16.1	-46.4	17.3	-47.4	18.3	-48.7	19.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-14.1	-29.3	-31.3	-29.0	-20.5	-7.8	2.9	6.2	4.0	-2.1	-13.8	-22.0	-27.6	(6)
(7)		DBStd	14.77	6.31	6.56	6.36	6.09	5.41	3.88	3.97	3.68	4.11	6.40	6.23	6.49	(7)
(8)		HDD10.0	8816	1219	1155	1209	915	553	218	133	191	362	737	958	1165	(8)
(9)		HDD18.3	11832	1478	1388	1468	1165	811	464	375	444	612	996	1208	1423	(9)
(10)		CDD10.0	27	0	0	0	0	0	4	17	6	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.5	4.7	4.1	4.3	4.2	4.6	5.2	5.3	4.8	4.9	4.2	3.7	4.1	(14)
(15)	Precipitation	PrecAvg	151	6	6	7	8	10	13	27	28	19	14	9	6	(15)
(16)		PrecMax	206	15	22	15	21	25	30	47	55	35	37	22	26	(16)
(17)		PrecMin	89	1	1	1	1	2	2	9	6	5	2	1	1	(17)
(18)		PrecStd	34	3	4	3	5	6	7	11	12	8	8	5	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.8	-11.8	-10.5	-0.8	5.5	18.8	20.7	20.3	9.2	0.5	-3.4	-9.5	(19)
(20)			MCWB	-11.1	-12.2	-10.9	-1.7	3.4	13.2	14.1	14.7	7.6	0.4	-3.8	-9.9	(20)
(21)		2%	DB	-15.4	-16.3	-14.4	-6.0	2.9	14.5	18.4	15.1	6.7	-1.3	-8.0	-12.6	(21)
(22)			MCWB	-15.9	-16.6	-14.8	-6.5	1.5	10.9	13.4	11.8	6.0	-1.6	-8.7	-12.9	(22)
(23)		5%	DB	-17.9	-18.7	-17.2	-8.6	1.2	11.0	16.0	11.9	4.8	-2.8	-11.2	-15.5	(23)
(24)			MCWB	-18.2	-19.1	-17.6	-9.1	0.2	8.3	11.8	9.8	4.3	-3.3	-11.5	-15.8	(24)
(25)		10%	DB	-20.0	-21.1	-19.8	-11.0	-0.3	8.6	13.5	9.6	3.0	-4.8	-13.4	-18.4	(25)
(26)			MCWB	-20.3	-21.4	-20.2	-11.4	-1.0	6.6	10.7	8.3	2.6	-5.2	-13.7	-18.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.0	-12.2	-10.8	-1.6	3.6	13.4	15.6	15.1	7.7	0.5	-3.7	-9.7	(27)
(28)			MCDWB	-10.7	-11.7	-10.4	-0.7	5.2	18.0	18.6	19.5	8.6	0.5	-3.3	-9.4	(28)
(29)		2%	WB	-15.8	-16.5	-14.8	-6.5	1.7	10.9	13.8	12.3	6.0	-1.5	-8.4	-12.8	(29)
(30)			MCDWB	-15.4	-16.2	-14.5	-6.1	2.7	14.3	17.7	15.1	6.7	-1.4	-8.0	-12.4	(30)
(31)		5%	WB	-18.2	-19.0	-17.5	-9.0	0.3	8.6	12.3	9.9	4.3	-3.2	-11.5	-15.8	(31)
(32)			MCDWB	-17.9	-18.7	-17.2	-8.5	1.1	10.9	15.9	11.7	4.8	-2.9	-11.2	-15.6	(32)
(33)		10%	WB	-20.3	-21.4	-20.1	-11.5	-1.0	6.6	10.6	8.3	2.7	-5.2	-13.7	-18.7	(33)
(34)			MCDWB	-20.0	-21.1	-19.8	-11.0	-0.2	8.4	13.4	9.5	3.0	-4.9	-13.5	-18.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.1	7.1	7.6	8.8	6.4	5.0	6.4	5.6	3.9	6.1	7.1	6.7	(35)
(36)			MCDBR	9.5	8.8	9.4	8.6	6.8	8.9	10.7	9.9	5.6	5.1	7.8	7.9	(36)
(37)			MCWBR	9.4	8.7	9.1	8.1	5.8	6.1	6.7	6.9	4.8	5.1	7.7	7.8	(37)
(38)		5% WB	MCDBR	9.4	8.9	9.2	8.4	6.2	8.9	10.5	9.1	5.4	5.1	8.0	7.8	(38)
(39)			MCWBR	9.4	8.8	9.0	8.0	5.3	6.3	7.1	6.7	4.8	5.1	8.0	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	N/A	0.164	0.237	0.308	0.326	0.332	0.330	0.321	0.277	0.168	N/A	N/A	(40)	
(41)		taud	N/A	1.879	2.133	2.014	2.063	2.288	2.402	2.419	2.345	1.923	N/A	N/A	(41)	
(42)		Ebn at Noon	0	350	702	765	814	834	824	774	666	310	0	0	(42)	
(43)		Edh at Noon	0	22	61	109	122	102	88	74	52	18	0	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.00	0.13	1.40	4.07	6.15	6.50	5.30	2.98	1.25	0.29	0.00	0.00	(44)	
(45)		RadStd	0.00	0.01	0.07	0.20	0.37	0.59	0.39	0.33	0.15	0.04	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page